

Name **KEY**

GT ID **Evans**

CHEM 2312 LS

Organic Chemistry II

Midterm Examination 1

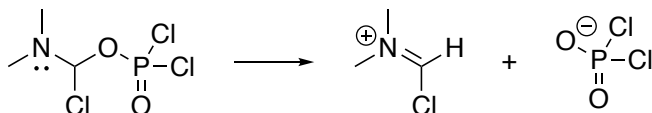
Take a deep breath. You got this.

Do not open this exam until you are instructed to do so. Only a pencil or pen, scratch paper, and molecular model kit are permitted. You will have 50 minutes to complete the exam; it can be submitted at any time after 25 minutes have elapsed. Write all responses clearly in the boxes or spaces provided; work outside of designated areas will not be graded. On multiple-choice questions, write the letter of your choice in the box *and* circle your choice.

Ensure that configurations can be unambiguously assigned at all stereocenters (*R/S* or *E/Z*). Use “+ En” to indicate the presence of a racemic mixture when chiral products are generated from achiral starting materials.

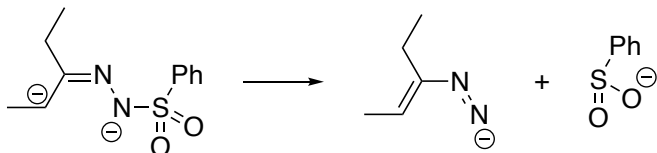
Multiple-choice Questions. Select the best response. Circle your choice and write its letter in the box. Each question is worth **4 points**.

1. Which label best describes the elementary step depicted below?



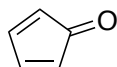
- B**
- A. Bimolecular nucleophilic substitution (S_N2)
 - B. Beta-elimination (E_β)**
 - C. 1,2-Rearrangement ($1,2R$)
 - D. Loss of a leaving group (D_N)
 - E. Bimolecular elimination ($E2$)

2. Which set of curved arrows correctly depicts the elementary step shown below?



- E**
- A** **B** **C** **D** **E**

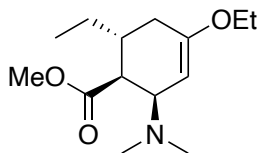
3. Cyclopentadienone (CPD) is a remarkably unstable compound that has only been prepared under extreme conditions. Which statement *best* explains why?



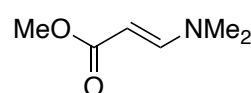
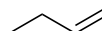
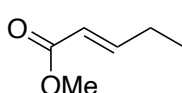
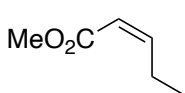
cyclopentadienone

- A**
- A. CPD has a significant alternative resonance form containing a fully conjugated ring with four pi electrons.**
 - B. The conjugated diene built into CPD is locked in the s-cis conformation, resulting in very rapid dimerization via (4 + 2) cycloaddition.
 - C. The fully conjugated five-membered ring exhibits very high strain.
 - D. The ring is forced into a nonplanar conformation and as a result is not actually conjugated.
 - E. The two alkene groups are electron withdrawing, making the carbonyl group extremely electrophilic.

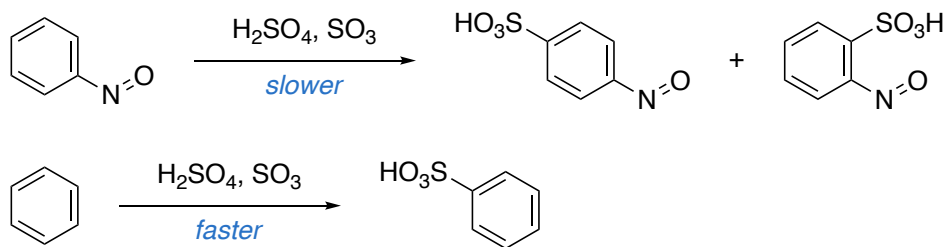
4. What is the structure of the dienophile needed to prepare the following compound using a Diels-Alder cycloaddition reaction?



C



5. The nitroso group ($-\text{NO}$) is an *ortho/para* director but nitrosobenzene reacts more slowly than benzene with sulfuric acid and sulfur trioxide. Which statement best explains this apparent discrepancy?



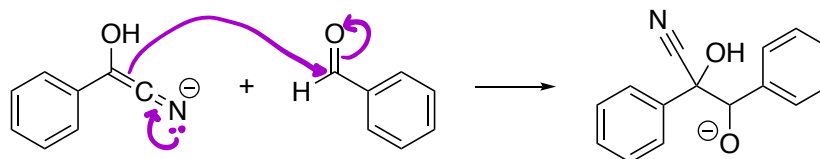
B

- A. The species that reacts with SO_3H^+ is *protonated* nitrosobenzene, not nitrosobenzene itself.
B. The nitroso group is electron donating by resonance but withdrawing by induction.
 C. Steric hindrance in nitrosobenzene prevents substitution at the *meta* position.
 D. The nitroso group forms a Lewis acid-base complex with SO_3 .
 E. [There is no discrepancy; *ortho/para* directing groups are typically deactivating toward electrophilic aromatic substitution.]

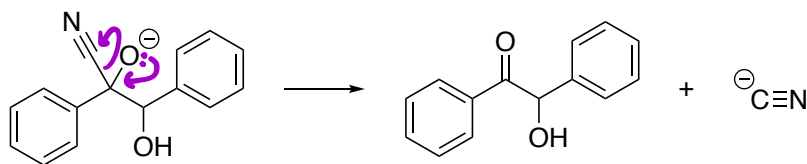
Other Questions. Place your response in the designated space for each question.

6. Each of the transformations below occurs in a single elementary step. Add curved arrows and any necessary implied hydrogens to the reactants below to show the flow of electrons in each step. Lone pairs are optional.

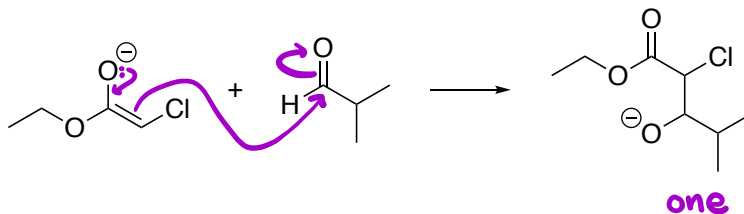
A.



B.

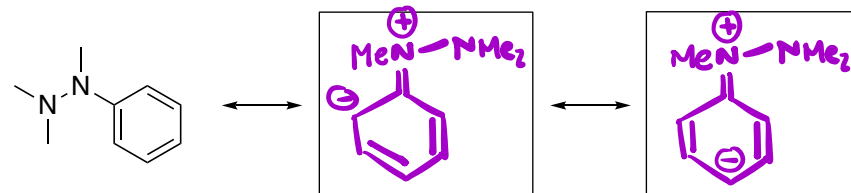


C.

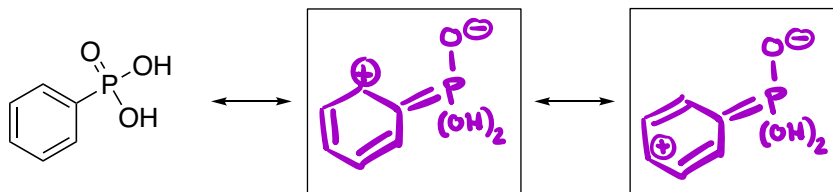


7. Provide two significant alternative resonance forms *containing formal charge in the benzene ring* for each of the compounds below. Ensure that each resonance form contains formal charges—don't just cycle the pi electrons in the ring.

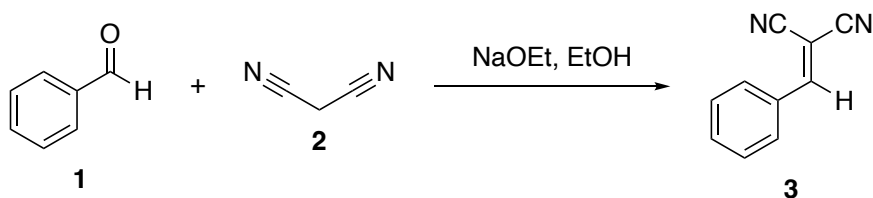
A.



B.



8. Compound **1** condenses with **2** in the presence of base to form product **3**.



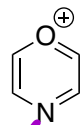
Which of the three compounds has the longest wavelength of maximum absorption? Check one box below and explain how you know.

☐ **1** ☐ **2** ☒ **3**

Explanation

Compound **3** has the longest conjugated system. The condensation reaction puts the phenyl ring of **1** in conjugation with the cyano groups of **2**.

9. Indicate whether each of the compounds below is aromatic, antiaromatic, or non-aromatic. Assume all structures are planar.



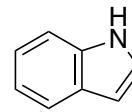
☒ Aromatic
☐ Antiaromatic
☐ Non-aromatic



☒ Aromatic
☐ Antiaromatic
☐ Non-aromatic



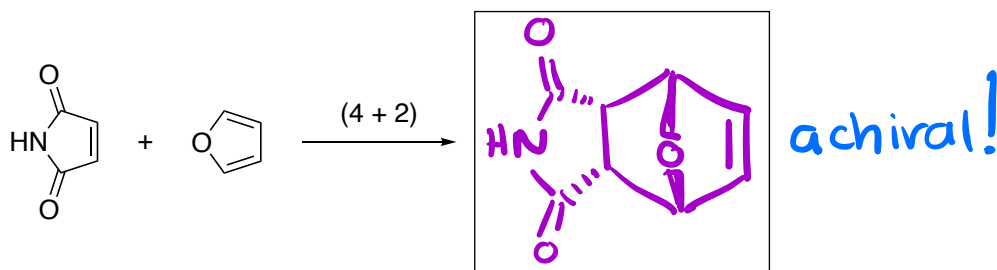
☒ Aromatic
☐ Antiaromatic
☐ Non-aromatic



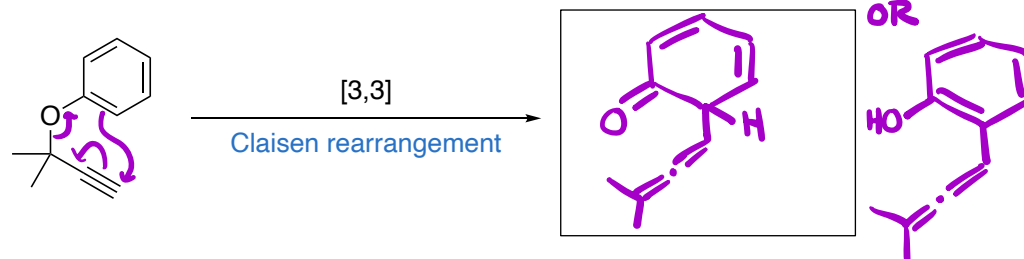
☒ Aromatic
☐ Antiaromatic
☐ Non-aromatic

10. Predict the neutral organic product of each reaction below. If multiple isomers are possible, draw only the major product.

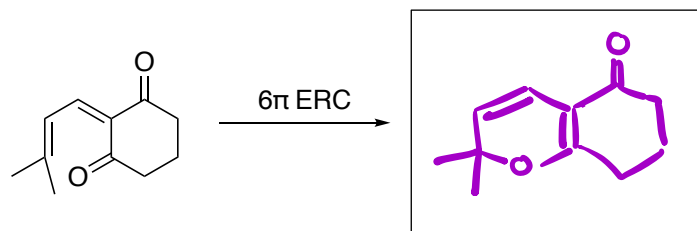
A.



B.

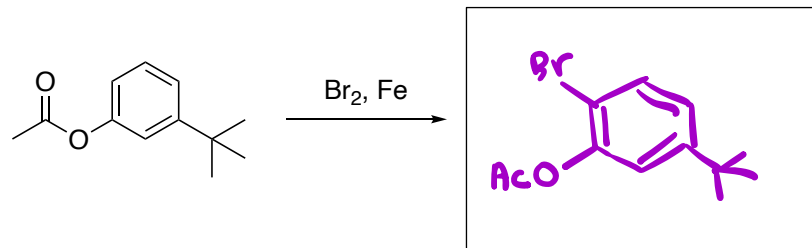


C.

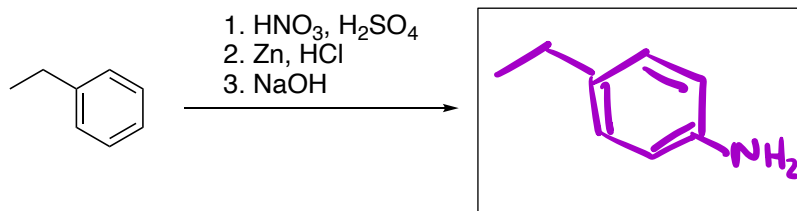


11. Predict the neutral organic product of each reaction below. If multiple isomers are possible, draw only the major product.

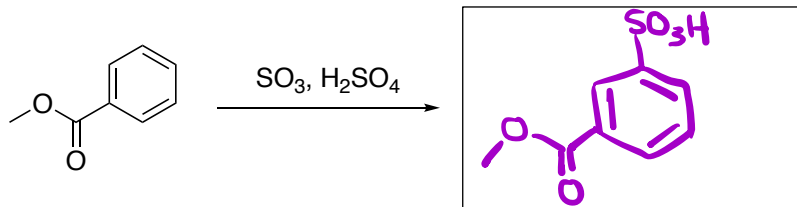
A.



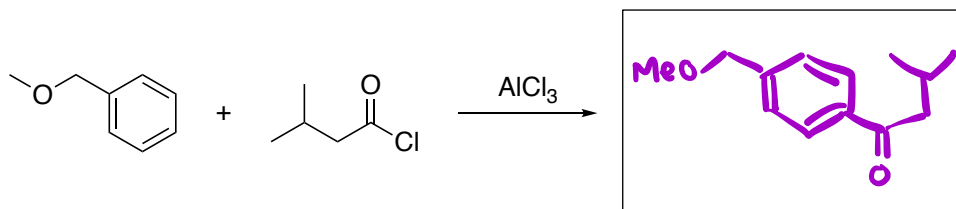
B.



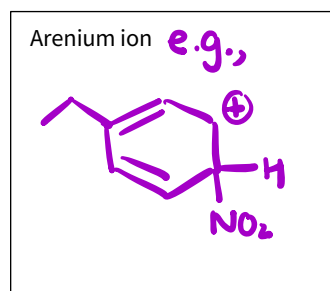
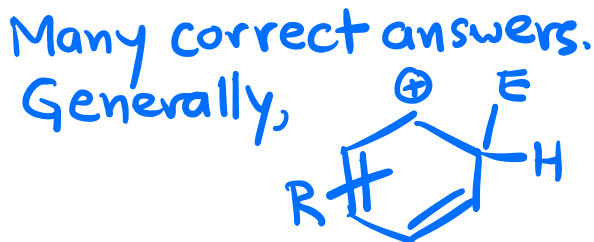
C.



D.

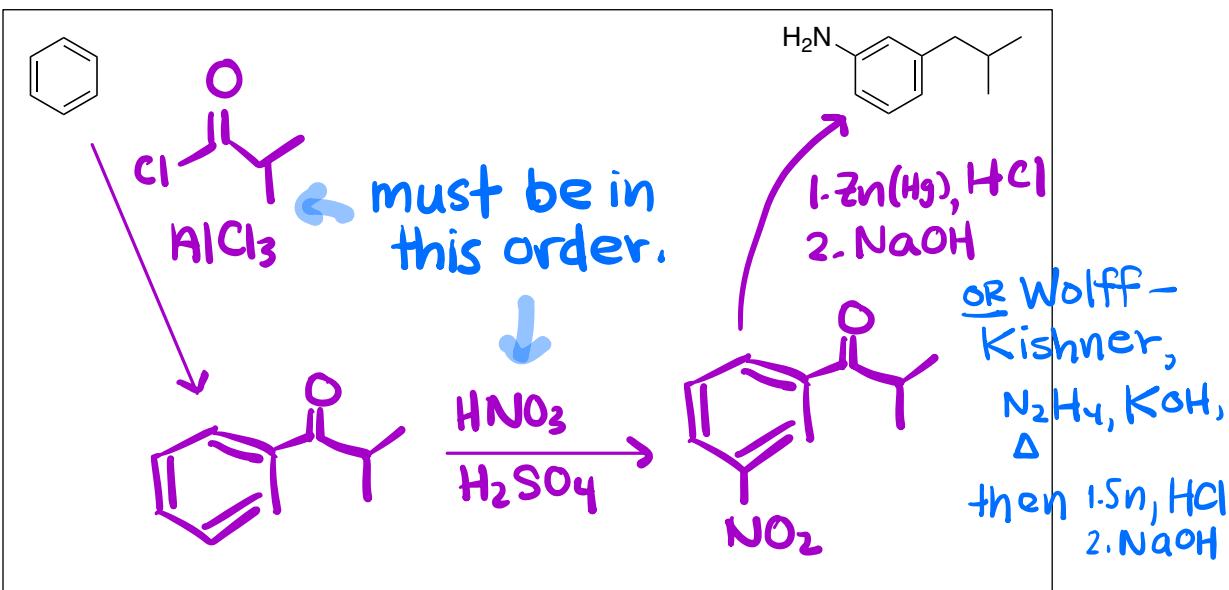


12. Provide the structure of an arenium intermediate involved in the mechanism of one of the reactions in problem 11 above.

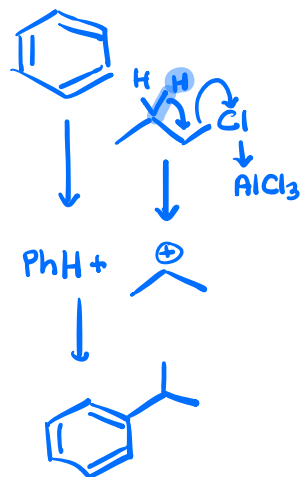


13. Show how each of the compounds below can be synthesized from benzene using a sequence of reactions. Include all reagents and structures for all isolated products in a synthetic scheme. Do not include mechanisms or reactive intermediates.

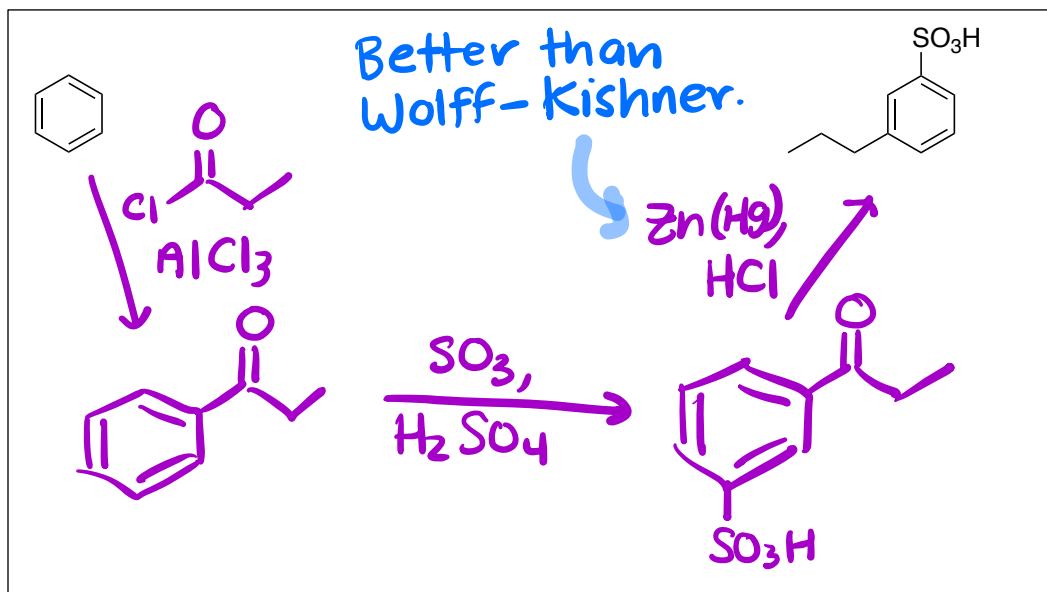
A.



B.



Don't alkylate w/
Primary RX. ☹



C.

